AP Precalculus **Unit 1, Lesson 1.14**

Warm-Up — Answer Key

Name: ____________________________ Date: ________________ Period: _________

1. A table of values for $f(x)$ is given below (where x is in seconds and $f(x)$ is in feet).

x	2	10
$f(x)$	8	32

Compute the average rate of change of f on the interval $[2, 10]$. Include appropriate units.

$$\text{ARC} = \frac{f(10) - f(2)}{10 - 2} = \frac{\mathbf{24}}{\mathbf{8}} = \mathbf{3}$$

Units: **feet per second (ft/s)**

2. Evaluate $p(x) = -3x^3 + 2x^2 - x + 5$ at $x = -2$. Show each step.

$p(-2) = -3(-\mathbf{2})^3 + 2(-\mathbf{2})^2 - (-\mathbf{2}) + 5 = -3(-8) + 2(4) + 2 + 5 = 24 + 8 + 2 + 5 = 39$

3. A scientist says: "As the distance between two magnets doubles, the force between them drops to one-fourth of its original value." Circle the model type that best fits this description and explain in one sentence.

A. Linear **B.** Quadratic **C.** **Rational (inverse proportion)** **D.** Cubic

Explanation: **Force $\propto 1/d^2$; doubling d multiplies d^2 by 4, so force becomes $1/4$ — this is an inverse-square (rational) model.**